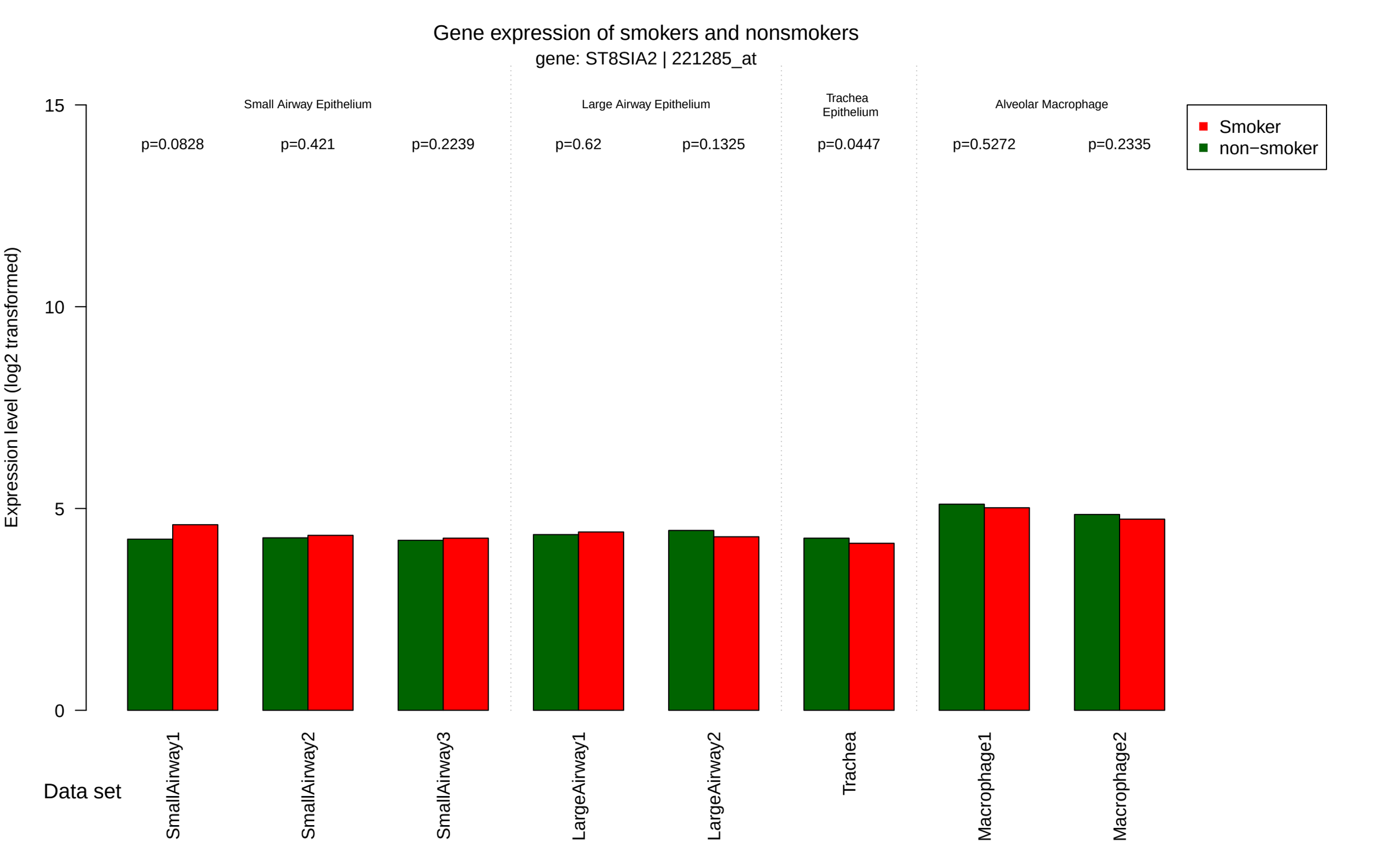
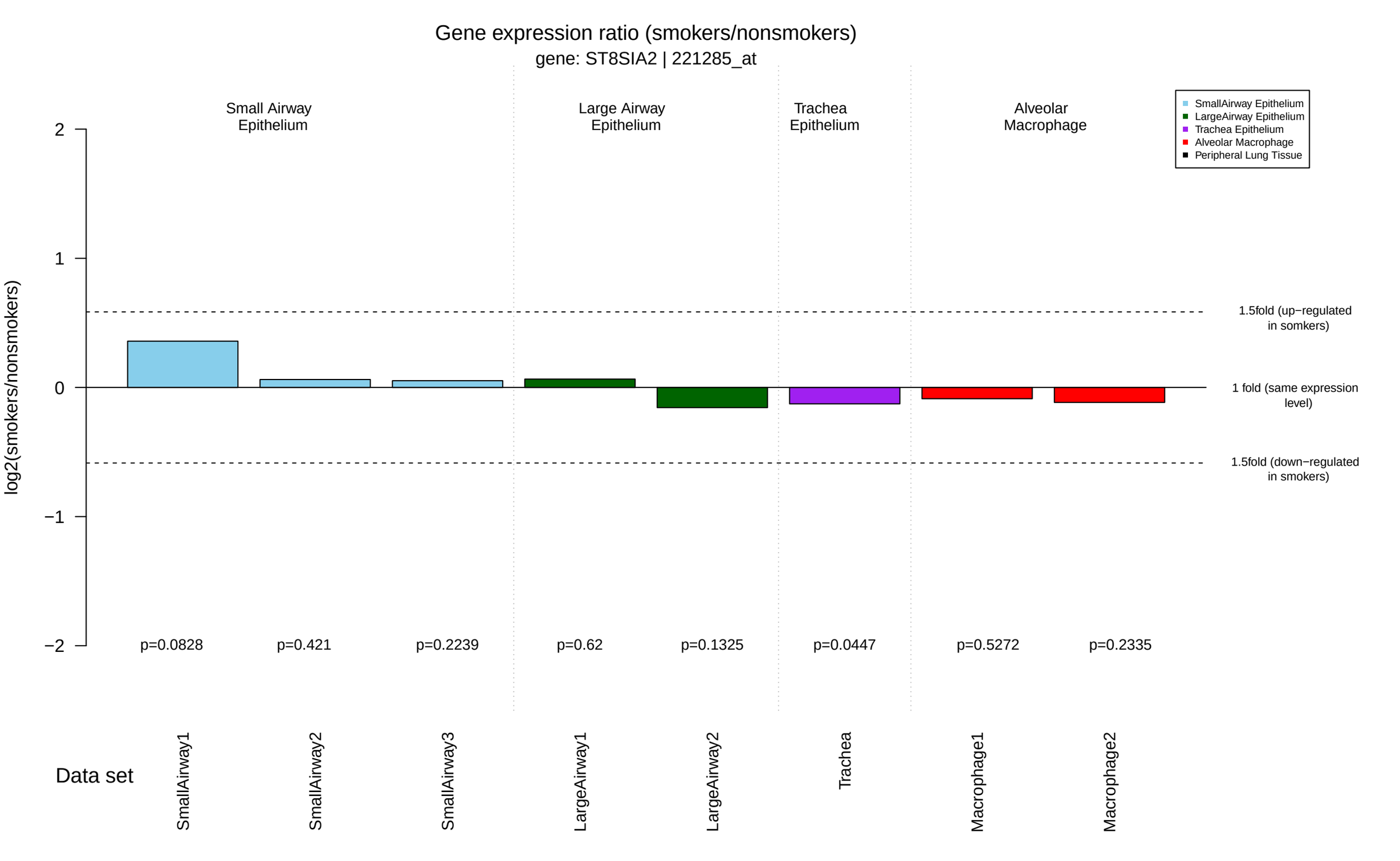
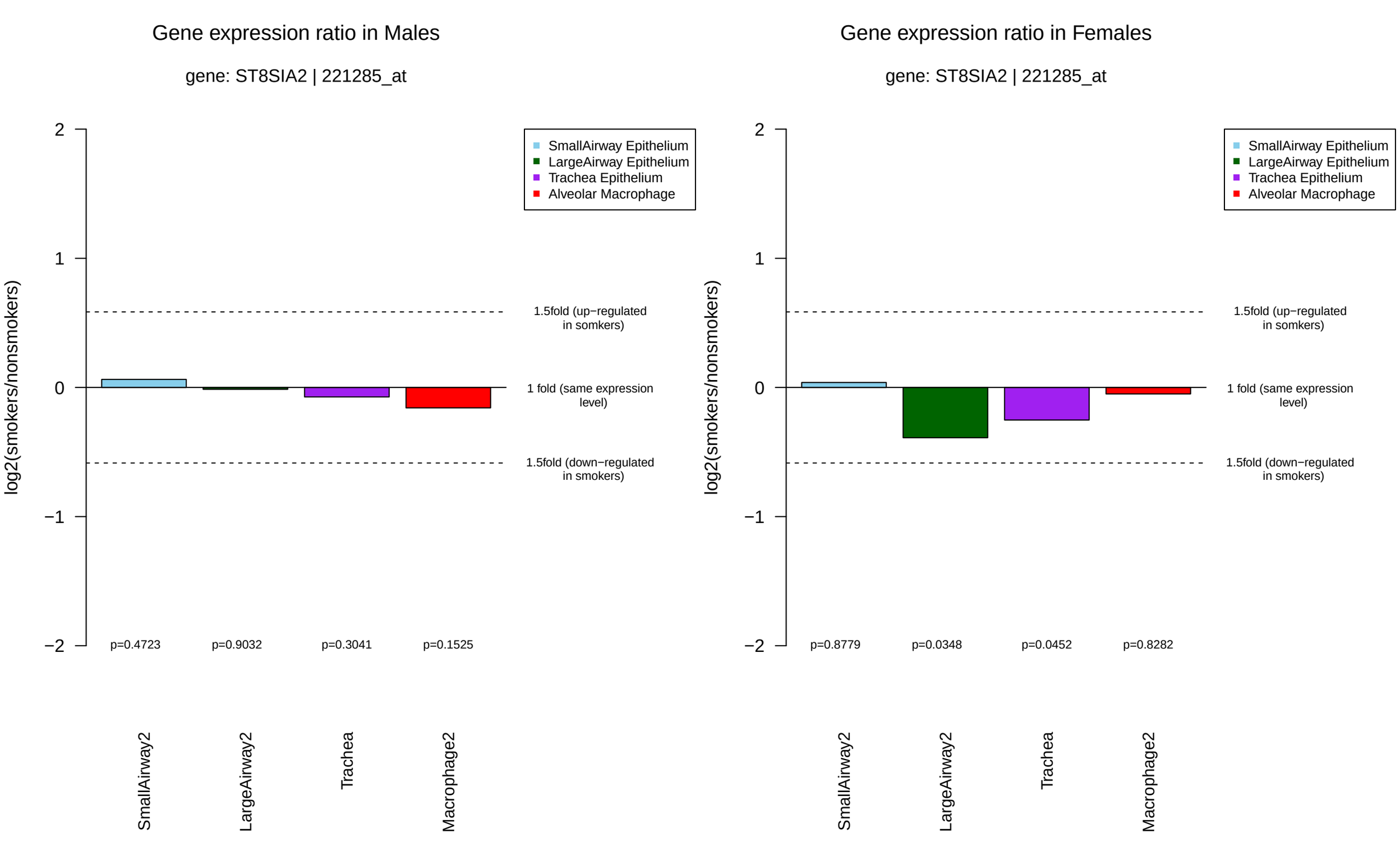
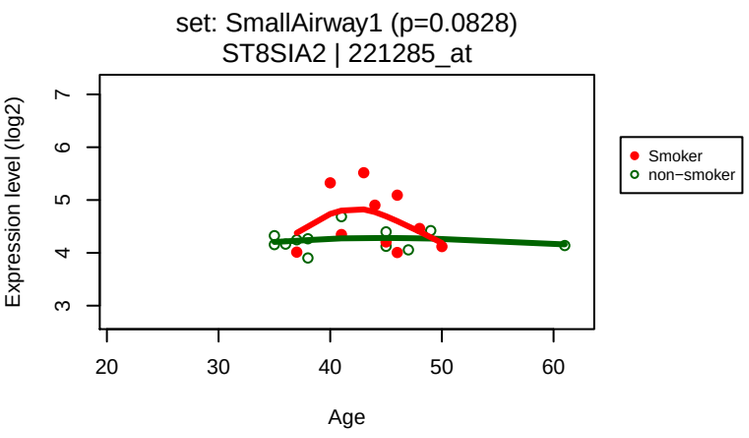
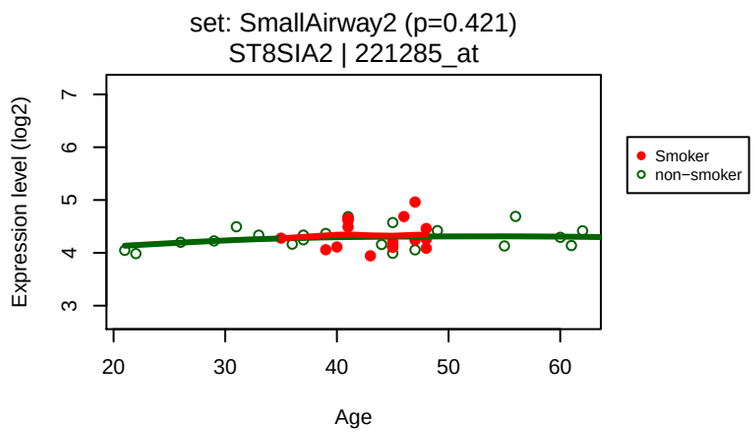
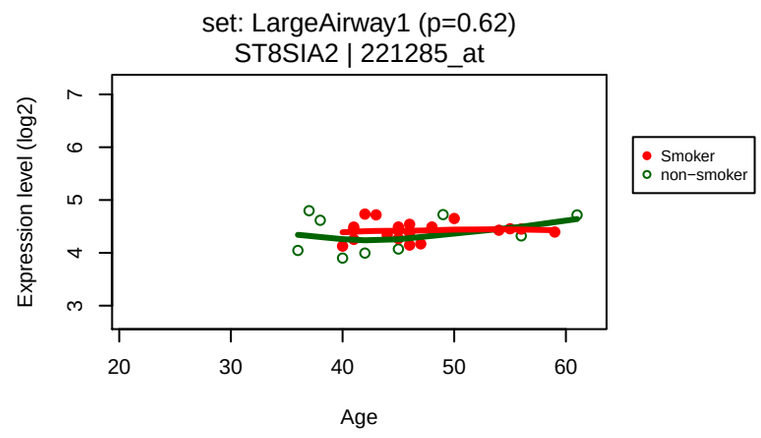

Gene expression of smokers and nonsmokers
gene: ST8SIA2 | 221285_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0828
p=0.421
p=0.2239
p=0.62
p=0.1325
p=0.0447
p=0.5272
p=0.2335
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

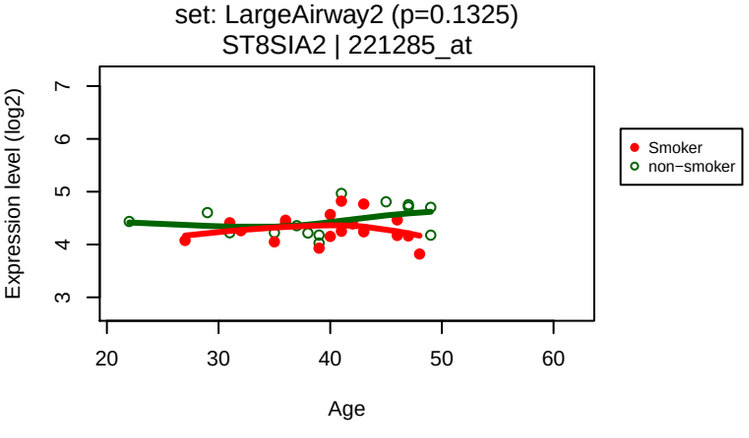

Gene expression ratio (smokers/nonsmokers)
gene: ST8SIA2 | 221285_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0828
p=0.421
p=0.2239
p=0.62
p=0.1325
p=0.0447
p=0.5272
p=0.2335
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

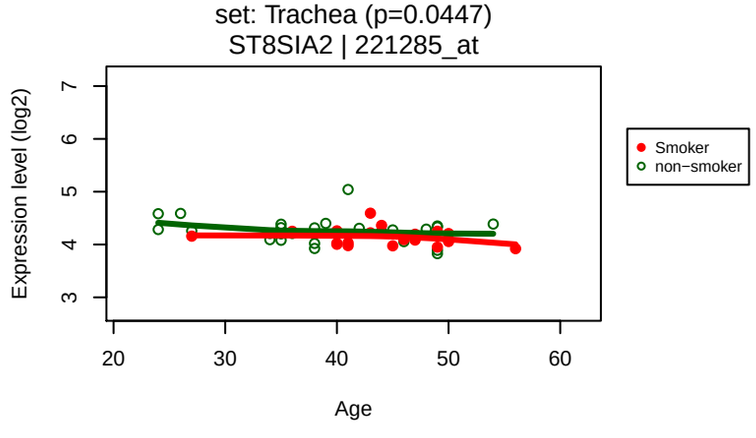

Gene expression ratio in Males
gene: ST8SIA2 | 221285_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4723
p=0.9032
p=0.3041
p=0.1525
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST8SIA2 | 221285_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8779
p=0.0348
p=0.0452
p=0.8282
SmallAirway2
LargeAirway2
Trachea
Macrophage2

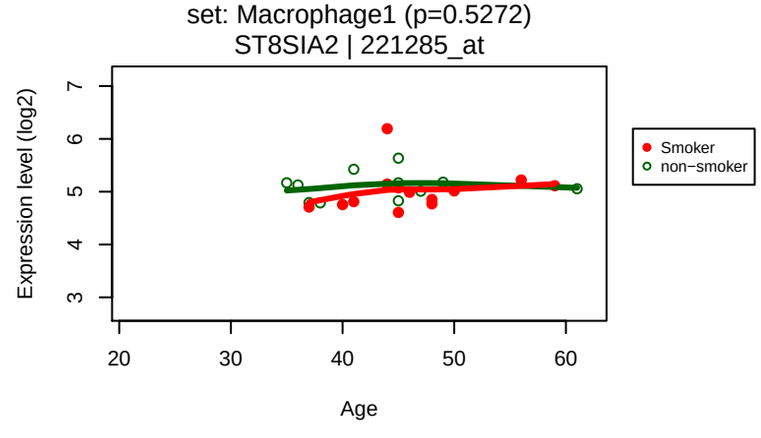

set: SmallAirway1 (p=0.0828)
ST8SIA2 | 221285_at
Expression level (log2)
Age
Smoker
non−smoker

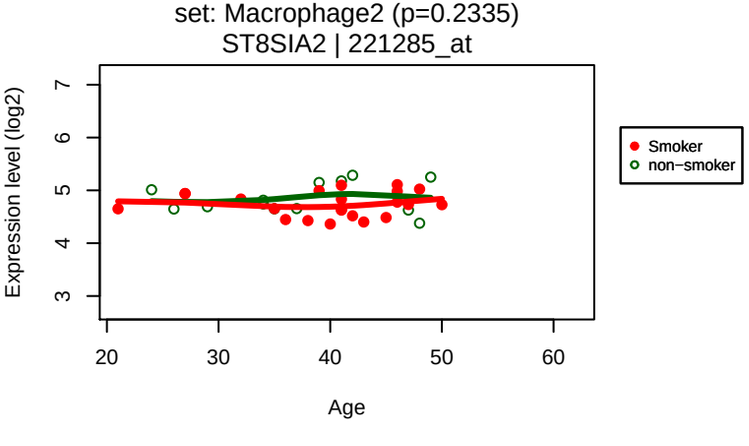

set: SmallAirway2 (p=0.421)
ST8SIA2 | 221285_at
Expression level (log2)
Age
Smoker
non−smoker

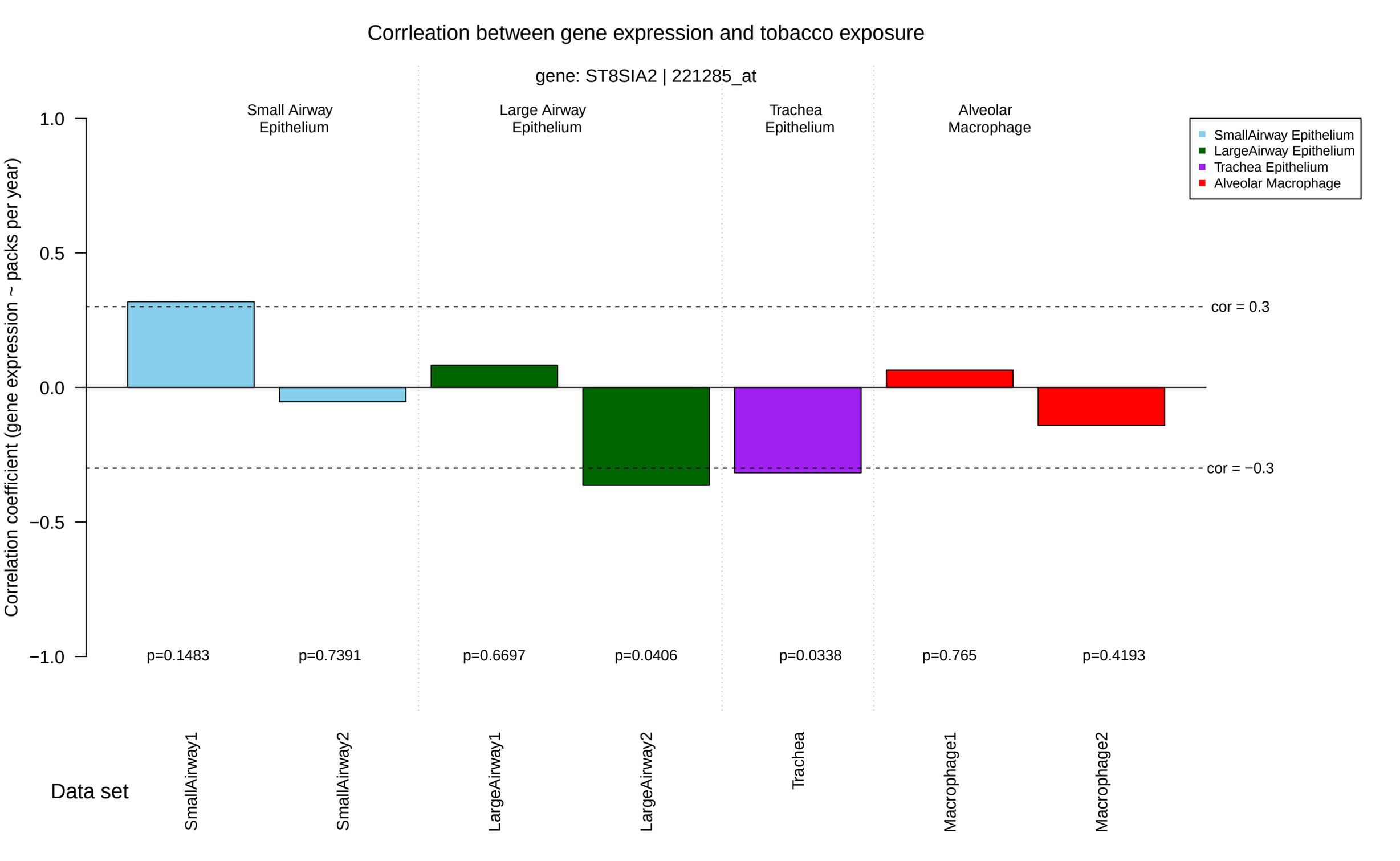

set: LargeAirway1 (p=0.62)
ST8SIA2 | 221285_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1325)
ST8SIA2 | 221285_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0447)
ST8SIA2 | 221285_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5272)
ST8SIA2 | 221285_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2335)
ST8SIA2 | 221285_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST8SIA2 | 221285_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1483
p=0.7391
p=0.6697
p=0.0406
p=0.0338
p=0.765
p=0.4193
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2